① Second-order Taylor polynomial

$$T_2(x,y) = f(x_0,y_0) + \nabla f(x_0,y_0)\cdot(x-x_0,\, y-y_0) + \frac{1}{2}(x-x_0,\, y-y_0)H(x_0,y_0)\begin{pmatrix}x-x_0\\ y-y_0\end{pmatrix}$$

a) $f(x,y) = \sin(x+2y)$ at $(0,0)$.

$f(0,0) = \sin(0+0) = 0.$

$\nabla f(x_0,y_0) = (1,2)$

$$\frac{\partial f}{\partial x} = \cos(x+2y) \qquad \frac{\partial^2 f}{\partial x^2} = -\sin(x+2y) \qquad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -2\sin(x+2y)$$

$$\frac{\partial f}{\partial y} = 2\cos(x+2y) \qquad \frac{\partial f}{\partial y^2} = -4\sin(x+2y)$$

$$H(0,0) = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$T_2(0,0) = 0 + (1,2)\cdot(x,y) + \underbrace{\frac{1}{2}\cdot\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}\cdot\begin{pmatrix}x-x_0\\ y-y_0\end{pmatrix}_{2\times 1}}_{0} = x+2y.$$

?
$t = x+2y$

$\Rightarrow \sin t = t - \frac{t^3}{3!} + \dots$

but $T_2 = t$

b) $f(x,y) = e^{\overbrace{x+y}^{t}}$ at $(0,0)$ and $(1,-1)$

Let $t = x+y \Rightarrow f(t) = e^t = \sum_{m \geq 0} \frac{t^m}{m!} = 1 + t + \frac{t^2}{2!} + \dots = \exp(t)$

$\Rightarrow T_2 = 1 + t + \frac{t^2}{2} = 1 + x + y + \frac{(x+y)^2}{2}$

$f(0,0) = 1$

$f(1,-1) = 1$

$\nabla f(x,y) =$

$$\frac{\partial f}{\partial x} = e^{x+y} = \frac{\partial f}{\partial y}$$

$$\frac{\partial^2 f}{\partial x^2} = e^{x+y} = \frac{\partial^2 f}{\partial^2 y} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

$$\Rightarrow \nabla f(0,0) = \left(\frac{\partial}{\partial x_1} \overset{?}{;} \frac{\partial}{\partial x_2}\right) = (1,1)$$

$$\nabla f(1,-1) = (1,1)$$

$$H(x,y) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$T_2(0,0) = 1 + (1,1)\cdot(x,y) + \frac{1}{2}(x,y)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= 1 + x + y + \frac{1}{2}(x,y)\cdot\begin{pmatrix} x+y \\ x+y \end{pmatrix}$$

$$= 1 + x + y + \frac{(x+y)^2}{2}$$

$$T_2(1,-1) = 1 + (1,1)(x+1, y-1) + \frac{1}{2}\cdot(x+1, y-1)\cdot\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x-1 \\ y+1 \end{pmatrix}$$

$$= 1 + x + \cancel{1} + y - \cancel{1} + \frac{1}{2}(x+1, y-1)\,(x - \cancel{1} + y + \cancel{1},\ x - \cancel{1} + y + \cancel{1})$$

$$= 1 + x + y + \frac{1}{2}(x+1, y-1)\cdot(x+y,\ x+y)$$

$$= 1 + x + y + \frac{(x+y)^2}{2} \qquad ?$$

② a) $f(x,y) = (y-1)e^x + (x-1)e^y$ at $(0,0)$.

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left((y-1)e^x + e^y\right) = (y-1)e^x.$$

$$\frac{\partial f}{\partial x} = (y-1)e^x + e^y$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(e^x + (x-1)e^y\right) = (x-1)e^y$$

$$\frac{\partial f}{\partial y} = e^x + (x-1)e^y$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial y}\left((y-1)e^x + e^y\right) = e^x + e^y = \frac{\partial^2 f}{\partial y \partial x}$$

$$\Rightarrow H(0,0) = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$$

eigenvalues: $A\sigma = \lambda\sigma$

$(A - \lambda I)\cdot\sigma = 0, \quad \sigma \neq 0$

$\det(A - \lambda I) = 0.$

$\det(H(0,0) - \lambda I) = 0$

$$\begin{vmatrix} -1-\lambda & 2 \\ 2 & -1-\lambda \end{vmatrix} = (-1-\lambda)^2 - 4 = \lambda^2 + 2\lambda - 3 = 0$$

$\Delta = 16 \rightarrow \lambda_1 = -3$ | eigenvalues

$\rightarrow \lambda_2 = 1.$

③ a) $f(x,y) = x^3 - 3x + y^2$

$$\frac{\partial f}{\partial x} = 3x^2 - 3 = 0 \Rightarrow 3x^2 = 3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1.$$

$$\frac{\partial f}{\partial y} = 2y = 0 \Rightarrow y = 0.$$

$\Rightarrow (1,0), (-1,0)$ critical points.

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(3x^2 - 3\right) = 6x.$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(2y\right) = 2.$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\Rightarrow H(1,0) = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \left.\begin{array}{l} \lambda_1 = 6 > 0 \\ \lambda_2 = 2 > 0 \end{array}\right\} \Rightarrow (1,0) \text{ local max}$$

$$\Rightarrow H(-1,0) = \begin{bmatrix} -6 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \left.\begin{array}{l} \lambda_1 = -6 < 0 \\ \lambda_2 = 2 > 0 \end{array}\right\} \Rightarrow \text{saddle point.}$$

④ $f : \mathbb{R}^n \to \mathbb{R}, \quad f(x) = \frac{1}{2} x^T A x.$

⑤